The Science of Defecting and Current U.S. Politics

November 2, 2026 — the eve of the midterm elections

Introduction

A dilemma between two persons is a moral question. Between five it becomes a statistical one, and between a hundred and fifty million voters it has stopped being a dilemma and become a climate, an atmosphere of suspicion in which each decides not what to do but what the others will have done. The classical prisoner's dilemma (Rapoport and Chammah, 1965) stages betrayal as a temptation; its many-player descendants (Hamburger, 1973; Kollock, 1998) stage it as a forecast. Two people placed before a table of payoffs that rewards mutual trust will nearly all cooperate. Five people placed before the *same* table, in which a single defector strips every cooperator of the round, abandon cooperation, although the arithmetic of the table has not moved.

The crowd, I will argue, leaves the payoffs alone. What it changes is the probability that cooperation will be honored and, more decisively, the *perceived* probability. Where the margin between outcomes is thin, defection is priced by its simulation instead of its actual risk, and the simulation arrives first: the defection precedes the defectors, as for Baudrillard the map precedes the territory (Baudrillard, 1994). Part I builds the mathematics of the multiplied dilemma, Part II its variations, and Part III applies it to the United States on the eve of the 2026 midterms.

I. The Arithmetic of the Crowd

The game of two

Let each of n players choose to cooperate (C) or defect (D). If all cooperate, each receives R. A defector receives a safe payoff T with certainty. A cooperator receives S if *any* other player defects. The ordering $S < T < R$ makes this, strictly, an n-player stag hunt: no one is tempted by a prize larger than mutual trust, only by safety (Skyrms, 2004). It keeps the dilemma's moral grammar of safety against faith, and it is the form the problem takes in politics, where the coalition's reward is large and the cost of being its last loyal member is ruinous.*

If each other player defects independently with probability p, a cooperator expects

$$E[C] = R\,(1-p)^{n-1} + S\left[1 - (1-p)^{n-1}\right], \tag{1}$$

*The same structure underlies the weak-link coordination game, in which each payoff depends on the lowest effort chosen by anyone in the group (Van Huyck et al., 1990).

and cooperation is optimal whenever $E[C] \geq T$, that is, whenever

$$(1-p)^{n-1} \;\geq\; q^* = \frac{T-S}{R-S}. \tag{2}$$

The left side is the probability that the coalition holds; the right side is the confidence cooperation requires. Take $R = 10$, $T = 7$, $S = 0$, so $q^* = 0.7$, and $p = 0.01$. Between two players the coalition holds with probability 0.99. Even a player who imagines her partner far less reliable, $\hat{p} = 0.10$, still sees 0.90, comfortably above the bar.

The game of many

Solving (2) for n gives the largest group in which trust remains rational,

$$n^* = 1 + \frac{\ln q^*}{\ln(1-p)}. \tag{3}$$

At the true rate $p = 0.01$, $n^* \approx 36$. At the imagined rate $\hat{p} = 0.10$, $n^* \approx 4.4$, and at five players cooperation is already irrational (Table 1). Nothing in the table moved. The player's error was raised to the power $n-1$. Doubt, in a crowd, is exponentiated.

n	$(0.99)^{n-1}$	$(0.90)^{n-1}$	Rational at $\hat{p} = 0.10$
2	0.990	0.900	cooperate
4	0.970	0.729	cooperate
5	0.961	0.656	defect
10	0.914	0.387	defect
36	0.703	0.025	defect

Table 1. Probability the coalition holds; $q^* = 0.7$.

This is what it means for the "optimal solution" to change. Universal cooperation remains the payoff-dominant equilibrium at every n, but the risk-dominant equilibrium, the one a prudent player selects under uncertainty (Harsanyi and Selten, 1988), migrates to defection as the group grows. The best outcome survives as an image while the rational act abandons it.

II. Variations on Betrayal

Thresholds and the cascade

In a crowd, p is nothing other than everyone's estimate of everyone else. Granovetter (1978) modeled collective behavior as a distribution of personal thresholds: each acts once enough others are expected to. Let player i carry a confidence threshold q_i with distribution F, and let a small core p_0 defect for private reasons. Each defects when the coalition's survival falls below her threshold, so

$$p_{t+1} = p_0 + (1-p_0)\left[1 - F\Big((1-p_t)^{n-1}\Big)\right]. \tag{4}$$

With $p_0 = 0.01$ and thresholds normal around 0.7 (s.d. 0.1), the equilibrium defection rate is about 1.5 percent for groups of four or fewer. At $n = 5$ the iteration runs to total defection. The collapse is a cliff, and one additional player is enough to push the group over it. The laboratory confirms the cliff: in weak-link experiments, pairs coordinated efficiently while groups of fourteen to sixteen sank to the worst equilibrium within a few rounds (Van Huyck et al., 1990), though large groups can hold if they are *grown* slowly from small ones whose history newcomers can see (Weber, 2006).

Size as a carrier of belief

Olson held that large groups fail to supply common goods because each contribution vanishes in the total (Olson, 1965); Boyd and Richerson (1988) showed that reciprocity cannot sustain cooperation in sizable groups, because the only way to punish one defector is to defect on everyone. Yet size alone is not the culprit: holding the return to contribution fixed, larger groups in Isaac and Walker (1988) did not contribute less. Size works through belief. The crowd is a machine for manufacturing $\hat{p}$, and its counterweights are institutions that make cooperation predictable: altruistic punishment (Boyd et al., 2003), monitored commons (Ostrom, 1990), and in democracies the unwritten norm of forbearance (Levitsky and Ziblatt, 2018), whose abandonment is "constitutional hardball" (Tushnet, 2004).

The voter's illusion

Rational choice treats a vote as a lottery ticket on a tie (Downs, 1957; Riker and Ordeshook, 1968). Quattrone and Tversky (1984) showed that people instead choose actions *diagnostic* of good outcomes even when the actions do not cause them: a voter reasons that if she turns out, people like her will too. In a crowd the individual decides as a type, on behalf of everyone like her. The illusion that makes her vote feel powerful also makes her loyalty feel expensive.

III. The Precession of Defection

The margin as threshold

Let a coalition hold a share s of the electorate and lead by a margin m. If a fraction δ of it switches sides, the margin moves by $2\delta s$; if it abstains, by δs. The race flips at

$$\delta^*_{\text{switch}} = \frac{m}{2s}, \qquad \delta^*_{\text{abstain}} = \frac{m}{s}. \tag{5}$$

The final YouGov polls in Iowa and Kansas showed the Democratic candidates ahead by a single point (*last updated Nov. 1, 2026*; Dorn, 2026); the Nebraska average showed Republican Pete Ricketts leading independent Dan Osborn by 1.7 (*last updated Nov. 1, 2026*; Wikipedia contributors, 2026b). With $m = 0.01$ and $s = 0.5$, $\delta^*_{\text{switch}} = 1\%$: one

voter in a hundred. The cooperator of the dyad could tolerate a 30 percent chance of betrayal; the one-point coalition must deliver ninety-nine of every hundred. The close race is the crowd game with q^* pushed toward certainty.

The dividend

On September 9, 2026, at the Republican Midterm Convention in Dallas, President Trump promised every adult citizen a $5,000 "Trump dividend" if Republicans keep both chambers of Congress (Brannon, 2026), at an estimated cost near $1.3 trillion; the earlier $2,000 "tariff dividend" was never paid (Nairametrics, 2026; Axios, 2025). One attorney called the pledge lawful because it would go to everyone regardless of their vote (Euronews, 2026); a former Deputy Assistant Attorney General and coauthors argued it may violate the federal vote-buying statutes and, by conditioning a payment described as already owed, the coercion statute (Swartz et al., 2026). Whatever the courts decide, I treat the offer as a defection: it breaks the compact by which a vote is a judgment on the common good rather than a claim on a payment, the inalienability Hasen (2000) places at the root of vote-buying law.*

Because it is paid to everyone, no individual must defect to collect it. In the strict calculus it is nearly worthless: in the most competitive states, the chance that one vote decided the 2008 presidential election was about one in ten million (Gelman et al., 2012). It can work only through the voter's illusion. If the voter treats her choice as mirroring a fraction λ of her coalition, she perceives herself decisive with probability

$$\hat{D}(m) = \min\left\{1,\ \frac{2\lambda s}{m}\right\}. \tag{6}$$

Let π be the perceived chance the payment materializes and $B = \$5{,}000$; the perceived cost of loyalty is $x(m) = \hat{D}(m)\,\pi B$. With illustrative values $\lambda = 0.005$ (she imagines one in two hundred of her bloc reasons as she does), $s = 0.5$, and $\pi = 0.2$, a five-point race prices loyalty at $100 and a one-point race at $500. The actual expected cost is about a hundredth of a cent. That gap is the simulation by which the crowd enters the individual.

Let each coalition member carry a price of conscience c_i, uniform on $[0, \bar{c}]$. The share defecting is $\delta(m) = x(m)/\bar{c} = A/(\bar{c}\,m)$, with $A = 2\lambda s\pi B$, and the race flips by its own anticipation when $\delta(m) \geq m/2s$, below the critical margin

$$m_c = \sqrt{2sA/\bar{c}}. \tag{7}$$

With $\bar{c} = \$20{,}000$, $m_c \approx 1.6$ points (Figure 1). Above it, a race is robust: the betrayal its closeness invites is smaller than the betrayal required. Below it, closeness manufactures the defection that ends it; at one point,

*Defenders answer that every campaign promises material benefits and that *Brown v. Hartlage* (1982) protects promises of policy; critics reply that a naked cash sum conditioned on a partisan outcome is price, not policy. The label here is philosophical, not legal.

2.5 percent of the coalition defects against a threshold of 1 percent. The perceived risk of loyalty outruns the actual risk, and the race flips before the actual numbers would justify it. This is the precession of defection. The parameters are illustrative; what is structural is the opposition of terms, perceived cost rising as $1/m$ while required betrayal falls as m.

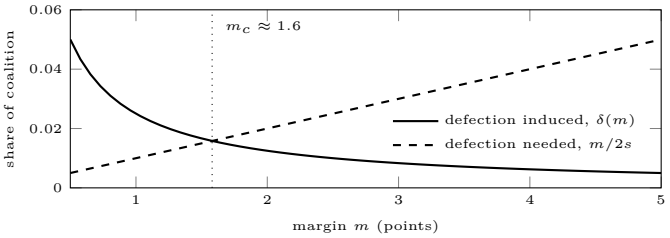


Figure 1. Below m_c, the defection induced exceeds the defection needed.

The national environment cut the other way: Democrats led the generic ballot by nearly nine points, the President's approval stood near 38 percent (*last updated Nov. 2, 2026*; Yahoo News, 2026), and betting markets gave Democrats about a nine-in-ten chance at the House (*last updated Nov. 1, 2026*; Dorn, 2026), all of which lowers any rational π. The model predicts no result. It locates where such an offer can bite, which is at the knife-edges: the races already within m_c.

The independent and the wasted vote

The same arithmetic governs loyalty *within* coalitions. In Nebraska the Democratic Party fielded no nominee and endorsed Osborn, who had lost to Senator Deb Fischer in 2024 by fewer than seven points (Wikipedia contributors, 2026b). His coalition is a pure stag hunt: it wins only if Democrats, independents, and dissenting Republicans all vote for the candidate they prefer, and each who suspects the others will revert to party faces the cooperator's S, a "wasted" vote, while the party line is the safe hare. Duverger's law (Duverger, 1954) is on this reading a belief equilibrium; third candidates die when voters expect them to (Myerson and Weber, 1993; Cox, 1997). Negative partisanship sweetens the hare, for when the other side's victory is the worst outcome imaginable, retreat to one's own side feels like prudence rather than betrayal (Abramowitz and Webster, 2016; Iyengar et al., 2019).

The map wars

The parties also defect against each other at the scale of states. In August 2025 Texas enacted a mid-decade map worth about five Republican seats; California answered with Proposition 50, approved by 64 percent of voters, worth about five Democratic seats. North Carolina, Ohio, Florida, Tennessee, Louisiana, and Alabama followed on one side, Utah's courts on the other, for a tally near sixteen Republican-favoring seats against six (Wikipedia contributors, 2026a). Here Boyd and Richerson's (1988) trap

is fully sprung: California could not punish Texas without abandoning the norm itself, so retaliation became indistinguishable from defection, as in the forty-three-day shutdown of 2025, a game of chicken in which each side was certain the other would swerve (Al Jazeera, 2025). One state declined. In December 2025 Indiana's Senate rejected a new map, twenty-one Republicans joining all ten Democrats (Wikipedia contributors, 2026a): Ostrom's monitors, who preferred the norm to the round, and a sign that the cliff depends on how many believe it is there.

Coda

The dilemma of two is a question of trust; the dilemma of the crowd is a question of the *image* of trust. Baudrillard described the masses as a body that absorbs meaning without returning it (Baudrillard, 1983). Here the mass absorbs risk and returns it as defection, raised to the power of its own size. The remedies follow from the mathematics: shrink $\hat{p}$ by making cooperation observable and coalitions remembered (Weber, 2006); design ballots on which a sincere vote is not a sucker's bet; and refuse prices offered to the collective, whose power lies wholly in the gap between what they are worth and what they appear to cost. Tomorrow's returns will not settle the theory. They will show only, in a handful of states, whether the defection arrived before the defectors.

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